

Problem 2

Part (a)

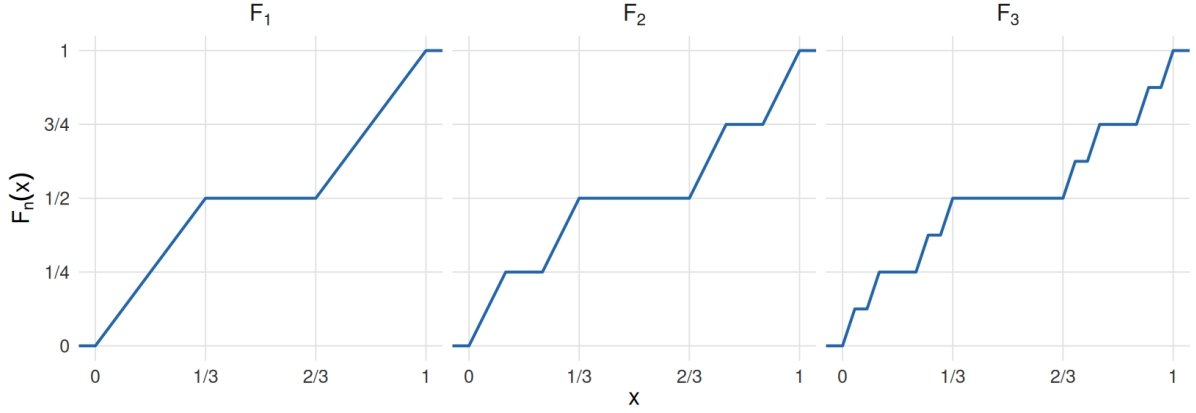


Figure 1: Graphs of F_1 , F_2 , and F_3 . Each n -th iteration introduces 2^{n-1} disjoint open intervals of length 3^{-n} inside $[0, 1]$ on which the function F_n is strictly constant (specifically, the intervals generated by the constant middle branch of the recurrence relation at the n -th iteration).

Part (b)

Show inequality ($|F_{n+1}(x) - F_n(x)| \leq \frac{1}{6 \cdot 2^n}, \forall x \in [0, 1], n \in \mathbb{N}_0$):

For $x \in [0, 1]$, we have that

$$F_0(x) = x$$

and

$$F_1(x) = \begin{cases} \frac{3}{2}x & \text{for } x \leq \frac{1}{3}, \\ \frac{1}{2} & \text{for } \frac{1}{3} < x < \frac{2}{3}, \\ \frac{3}{2}x - \frac{1}{2} & \text{for } \frac{2}{3} \leq x. \end{cases}$$

We show the inequality by induction over $n \in \mathbb{N}_0$:

Base Case ($n = 0$): We must show that $|F_1(x) - F_0(x)| \leq \frac{1}{6 \cdot 2^0} = \frac{1}{6}$ for $x \in [0, 1]$.

- For $x \in [0, \frac{1}{3}]$:

$$|F_1(x) - F_0(x)| = \left| \frac{3}{2}x - x \right| = \frac{1}{2}x \stackrel{(x \leq \frac{1}{3})}{\leq} \frac{1}{2} \left(\frac{1}{3} \right) = \frac{1}{6}.$$

- For $x \in (\frac{1}{3}, \frac{2}{3})$:

$$|F_1(x) - F_0(x)| = \left| \frac{1}{2} - x \right|.$$

Since $\frac{1}{3} < x < \frac{2}{3}$, it follows that $\frac{1}{2} - \frac{2}{3} = -\frac{1}{6} < \frac{1}{2} - x < \frac{1}{6} = \frac{1}{2} - \frac{1}{3}$, which implies $|\frac{1}{2} - x| < \frac{1}{6}$.

- For $x \in [\frac{2}{3}, 1]$:

$$|F_1(x) - F_0(x)| = \left| \frac{3}{2}x - \frac{1}{2} - x \right| = \left| \frac{1}{2}(1 - x) \right| \stackrel{(x \leq 1)}{=} \frac{1}{2}(1 - x) \stackrel{(x \geq \frac{2}{3})}{\leq} \frac{1}{2} \left(1 - \frac{2}{3} \right) = \frac{1}{6}.$$

Thus, the base case holds.

Inductive Step ($n \implies n + 1$): Assume the claim holds for an arbitrary $n \in \mathbb{N}_0$, i.e., $|F_{n+1}(x) - F_n(x)| \leq \frac{1}{6 \cdot 2^n}$ for all $x \in [0, 1]$. We evaluate the difference $|F_{(n+1)+1}(x) - F_{(n+1)}(x)| = |F_{n+2}(x) - F_{n+1}(x)|$:

- For $x \in [0, \frac{1}{3}]$:

$$|F_{n+2}(x) - F_{n+1}(x)| = \frac{1}{2} |F_{n+1}(3x) - F_n(3x)|.$$

Since $x \in [0, \frac{1}{3}]$, then $3x \in [0, 1]$ and we can apply the inductive hypothesis:

$$\frac{1}{2} |F_{n+1}(3x) - F_n(3x)| \leq \frac{1}{2} \cdot \frac{1}{6 \cdot 2^n} = \frac{1}{6 \cdot 2^{n+1}}.$$

- For $x \in (\frac{1}{3}, \frac{2}{3})$:

$$|F_{n+2}(x) - F_{n+1}(x)| = \left| \frac{1}{2} - \frac{1}{2} \right| = 0 \leq \frac{1}{6 \cdot 2^{n+1}}.$$

- For $x \in [\frac{2}{3}, 1]$:

$$|F_{n+2}(x) - F_{n+1}(x)| = \frac{1}{2} |F_{n+1}(3x - 2) - F_n(3x - 2)|.$$

Since $x \in [\frac{2}{3}, 1]$, then $3x - 2 \in [0, 1]$, and we can apply the inductive hypothesis:

$$\frac{1}{2} |F_{n+1}(3x - 2) - F_n(3x - 2)| \leq \frac{1}{2} \cdot \frac{1}{6 \cdot 2^n} = \frac{1}{6 \cdot 2^{n+1}}.$$

Therefore, the inequality holds for all $n \in \mathbb{N}_0$. □

Justification of well-definedness:

A limit function $F(x) := \lim_{n \rightarrow \infty} F_n(x)$ is well-defined if the sequence $(F_n(x))_{n \in \mathbb{N}_0}$ converges for every $x \in \mathbb{R}$. We distinguish three cases:

- For $x \in [0, 1]$, let $m > n$:

$$\begin{aligned} |F_m(x) - F_n(x)| &\leq \sum_{k=n}^{m-1} |F_{k+1}(x) - F_k(x)| \\ &\leq \sum_{k=n}^{\infty} \frac{1}{6 \cdot 2^k} = \frac{1}{6 \cdot 2^n} \sum_{j=0}^{\infty} \left(\frac{1}{2}\right)^j = \frac{1}{6 \cdot 2^n} \cdot 2 = \frac{1}{3 \cdot 2^n}. \end{aligned}$$

Since $\lim_{n \rightarrow \infty} \frac{1}{3 \cdot 2^n} = 0$, the sequence $(F_n(x))_{n \in \mathbb{N}_0}$ is a Cauchy sequence. Because $\mathbb{R}$ is complete, the sequence converges for all $x \in [0, 1]$.

- For $x < 0$: By induction, $F_0(x) = 0$ (base case). Assuming $F_n(y) = 0$ for all $y < 0$, it follows that for any $x < 0$, $3x < 0$, yielding $F_{n+1}(x) = \frac{1}{2} F_n(3x) = \frac{1}{2}(0) = 0$ (inductive step). Thus, $F_n(x) = 0$ for all $n \in \mathbb{N}_0$, implying $\lim_{n \rightarrow \infty} F_n(x) = 0$.
- For $x > 1$: By induction, $F_0(x) = 1$ (base case). Assuming $F_n(y) = 1$ for all $y > 1$, it follows that for any $x > 1$, $3x - 2 > 1$, yielding $F_{n+1}(x) = \frac{1}{2} F_n(3x - 2) + \frac{1}{2} = \frac{1}{2}(1) + \frac{1}{2} = 1$ (inductive step). Thus, $F_n(x) = 1$ for all $n \in \mathbb{N}_0$, implying $\lim_{n \rightarrow \infty} F_n(x) = 1$.

Consequently, the sequence $(F_n(x))_{n \in \mathbb{N}_0}$ converges for all $x \in \mathbb{R}$, and therefore the limit function $F(x)$ is well-defined. □

Part (c)

1. Show that F is a distribution function: To show that $F(x) := \lim_{n \rightarrow \infty} F_n(x)$ is a distribution function, we need to verify the following properties:

- **Limits at infinity:** In Part (b), we showed that for all $n \in \mathbb{N}_0$, $F_n(x) = 0$ for $x < 0$ and $F_n(x) = 1$ for $x > 1$. Therefore $\lim_{x \rightarrow -\infty} F(x) = \lim_{x \rightarrow -\infty} 0 = 0$ and $\lim_{x \rightarrow \infty} F(x) = \lim_{x \rightarrow \infty} 1 = 1$.
- **Continuity (and Right-Continuity):** We show by induction that F_n is continuous for all $n \in \mathbb{N}_0$.
 - *Base case ($n = 0$):* $F_0(x)$ is defined piecewise by continuous functions (0, x , and 1). At the boundary points, the limits and function value coincide: $\lim_{x \uparrow 0} F_0(x) = 0 = \lim_{x \downarrow 0} F_0(x) = F_0(0)$ and $\lim_{x \uparrow 1} F_0(x) = 1 = \lim_{x \downarrow 1} F_0(x) = F_0(1)$. Therefore, F_0 is continuous on $\mathbb{R}$.
 - *Inductive step ($n \implies n+1$):* Assume F_n is continuous. F_{n+1} consists of compositions of continuous functions ($F_n(3x)$ and $F_n(3x-2)$), making it continuous on $(-\infty, \frac{1}{3})$, $(\frac{1}{3}, \frac{2}{3})$, and $(\frac{2}{3}, \infty)$. We check the boundaries: $\lim_{x \uparrow 1/3} F_{n+1}(x) = \frac{1}{2}F_n(1) = \frac{1}{2} = \lim_{x \downarrow 1/3} F_{n+1}(x)$ and $\lim_{x \uparrow 2/3} F_{n+1}(x) = \frac{1}{2} = \frac{1}{2}F_n(0) + \frac{1}{2} = \lim_{x \downarrow 2/3} F_{n+1}(x)$. Thus, F_{n+1} is continuous on $\mathbb{R}$.

In Part (b), we showed the bound $|F_m(x) - F_n(x)| \leq \frac{1}{3 \cdot 2^n}$ for all $x \in \mathbb{R}$. Because this bound depends only on n and is independent of x , the sequence of functions $(F_n)_{n \in \mathbb{N}_0}$ is uniformly Cauchy, ensuring it converges uniformly to the limit function F . Since each F_n is continuous (shown above) and the uniform limit of continuous functions is continuous (Uniform Limit Theorem), then F is continuous on $\mathbb{R}$, which implies F is also right-continuous.

- **Monotonicity:** We first show by induction that F_n is monotonically increasing for all $n \in \mathbb{N}_0$.
 - *Base case ($n = 0$):* The function F_0 is constant (and thus non-decreasing) on $(-\infty, 0)$ and $(1, \infty)$, and, on the interval $[0, 1]$, the function is strictly increasing. Therefore, F_0 is monotonically increasing on $\mathbb{R}$.
 - *Inductive step ($n \implies n+1$):* Assume F_n is monotonically increasing (inductive hypothesis). We analyze F_{n+1} across its defined intervals and boundary points:
 - * **For $x \leq \frac{1}{3}$:** $F_{n+1}(x) = \frac{1}{2}F_n(3x)$. Since the inner function $3x$ is strictly increasing and F_n is monotonically increasing (by the inductive hypothesis), the composition $\frac{1}{2}F_n(3x)$ is monotonically increasing.
 - * **For $\frac{1}{3} < x < \frac{2}{3}$:** $F_{n+1}(x) = \frac{1}{2}$. This is constant, which satisfies the condition of being monotonically increasing (i.e., non-decreasing).
 - * **For $x \geq \frac{2}{3}$:** $F_{n+1}(x) = \frac{1}{2}F_n(3x-2) + \frac{1}{2}$. Since $3x-2$ is strictly increasing and F_n is monotonically increasing, the composition $\frac{1}{2}F_n(3x-2) + \frac{1}{2}$ is monotonically increasing.

Because F_{n+1} is monotonically increasing on every sub-interval and F_{n+1} is continuous everywhere (previous step), F_{n+1} is monotonically increasing on $\mathbb{R}$.

Then, we show that the limit function F is monotonically increasing. Let $x, y \in \mathbb{R}$ such that $x \leq y$. Since every F_n is monotonically increasing, we have $F_n(x) \leq F_n(y)$ for all $n \in \mathbb{N}_0$. Taking the limit as $n \rightarrow \infty$ and using the fact that limits preserve weak inequalities yields:

$$F(x) = \lim_{n \rightarrow \infty} F_n(x) \leq \lim_{n \rightarrow \infty} F_n(y) = F(y)$$

Hence, the limit function F is monotonically increasing.

Therefore, F satisfies the definition of a distribution function (Definition 1.4.6). By Theorem 1.4.5, F uniquely induces a probability measure $\mathbb{P}$ on the measurable space $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ such that $\mathbb{P}((-\infty, x]) = F(x)$ for all $x \in \mathbb{R}$. □

2. Point Probabilities:

For any probability measure $\mathbb{P}$ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$, the probability of an elementary event $\{x\}$ is given by the jump size of its distribution function:

$$\mathbb{P}(\{x\}) = F(x) - F(x-)$$

where $F(x-) = \lim_{y \uparrow x} F(y)$. Because we have established that F is continuous everywhere on $\mathbb{R}$, the left-sided limit evaluates exactly to the function value, yielding $F(x-) = F(x)$. Consequently:

$$\mathbb{P}(\{x\}) = F(x) - F(x) = 0 \quad \text{for all } x \in \mathbb{R}.$$
□

Part (d)

According to Definition 1.5.2, if $\mathbb{P}$ has a density, there exists a non-negative Borel-measurable function f such that $F(x) = \int_{-\infty}^x f(y)dy$ for all $x \in \mathbb{R}$. Consequently, for any Borel set $A \in \mathcal{B}(\mathbb{R})$, $\mathbb{P}(A) = \int_A f(y)dy$ (Theorem 1.5.4 (i)).

Proof by Contradiction: Assume that $\mathbb{P}$ has a density f :

1. **Probability measure of the constant intervals.** Let O_n denote the union of the 2^{n-1} disjoint open intervals of length 3^{-n} inside $[0, 1]$ on which the function F_n is strictly constant (specifically, the intervals generated by the constant middle branch of the recurrence relation at the n -th iteration, see Figure 1). On every open interval $I \subset O_n$, the limit function F is constant. Because F is continuous (Part c), for any such interval $I = (a, b)$, the point probability is $\mathbb{P}(\{b\}) = 0$. Thus, the probability measure is:

$$\mathbb{P}((a, b)) = \mathbb{P}((a, b]) - \mathbb{P}(\{b\}) = (F(b) - F(a)) - 0 = 0$$

Let $O = \bigcup_{n=1}^{\infty} O_n$ be the union of all such constant open intervals. Since O is a countable union of disjoint open intervals on which $\mathbb{P}$ assigns measure 0, it follows from the σ -additivity of $\mathbb{P}$ that:

$$\mathbb{P}(O) = \mathbb{P}\left(\bigcup_{n=1}^{\infty} O_n\right) = \sum_{n=1}^{\infty} \mathbb{P}(O_n) = 0$$

2. **Probability measure of the complement set.** We first prove that $F(0) = 0$ and $F(1) = 1$. From Part (b), for all $n \in \mathbb{N}_0$, $F_n(x) = 0$ for $x < 0$ and $F_n(x) = 1$ for $x > 1$, implying $F(x) = 0$ on $(-\infty, 0)$ and $F(x) = 1$ on $(1, \infty)$. Because F is continuous on all of $\mathbb{R}$ (Part c), its values at the boundaries equal its one-sided limits from these adjacent intervals: $F(0) = F(0-) = 0$ and $F(1) = F(1+) = 1$.

Let $C = [0, 1] \setminus O$ be the complement of these open intervals relative to $[0, 1]$. The probability measure of the closed interval is therefore $\mathbb{P}([0, 1]) = F(1) - F(0-) = 1 - 0 = 1$. Using the complement rule for probability measures:

$$\mathbb{P}(C) = \mathbb{P}([0, 1]) - \mathbb{P}(O) = 1 - 0 = 1$$

3. **Lebesgue measure of the complement set.** We now determine the Lebesgue measure λ of C . At each step $n \in \mathbb{N}$, the set O_n consists of 2^{n-1} intervals, each of length 3^{-n} . The total Lebesgue measure of O is the sum of these lengths:

$$\lambda(O) = \sum_{n=1}^{\infty} 2^{n-1} \left(\frac{1}{3}\right)^n = \frac{1}{3} \sum_{n=1}^{\infty} \left(\frac{2}{3}\right)^{n-1} = \frac{1}{3} \cdot \left(\frac{1}{1 - \frac{2}{3}}\right) = \frac{1}{3} \cdot 3 = 1$$

Since $\lambda([0, 1]) = 1$, the Lebesgue measure of C is:

$$\lambda(C) = \lambda([0, 1]) - \lambda(O) = 1 - 1 = 0$$

4. **Contradiction.**

Under our initial assumption that $\mathbb{P}$ has a density f , the probability of C must be:

$$\mathbb{P}(C) = \int_C f(y) dy$$

Because $\lambda(C) = 0$, the Lebesgue integral over C is strictly 0, implying $\mathbb{P}(C) = 0$. This directly contradicts our deduction in Step 2 that $\mathbb{P}(C) = 1$.

Therefore, our assumption must be false. There exists no non-negative, Borel-measurable function f such that $\int_{-\infty}^x f(y) dy = F(x)$ for all $x \in \mathbb{R}$. The probability measure $\mathbb{P}$ does not have a density.

□